

# RAVI KSHIRSAGAR

Generative AI Engineer · AI Application Developer · Prompt Engineer  
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## PROFESSIONAL SUMMARY

Final-year B.Tech (AI & ML) student with hands-on experience building production-oriented Generative AI applications, LLM-integrated agents, and Computer Vision systems. Proficient in Python, LangChain, RAG pipelines, OpenAI API, and prompt engineering (zero-shot, few-shot, chain-of-thought). Developed 4+ end-to-end AI projects demonstrating full-stack GenAI engineering across real-world use cases. Actively seeking AI/ML engineering internship roles to apply skills in a collaborative, fast-paced team environment.

## CORE TECHNICAL SKILLS

**Generative AI & LLMs:** Prompt Engineering (zero-shot, few-shot, CoT, role-based), LLM API Integration, AI Agent Development, RAG Pipelines, Semantic Search, Multimodal AI, NLP

**Frameworks & Libraries:** LangChain, ChromaDB, FAISS, Sentence Transformers, NumPy, Pandas, OpenCV, YOLOv8, Reinforcement Learning

**Languages:** Python, TypeScript, JavaScript, HTML5 / CSS3

**Tools & Platforms:** OpenAI API, Claude AI (Anthropic), Gemini / Google AI Studio, Firebase, Streamlit, Git / GitHub, Cursor AI, REST APIs

**Practices:** Agile / Scrum, Problem Solving, Debugging, Technical Documentation

## PROJECTS

**Career Pilot AI** | *TypeScript · Firebase · LLM APIs · GenAI*

[ In Development ]

- Engineered a full-stack GenAI career platform that automates ATS resume scoring and gap analysis using multi-model LLM APIs (Claude AI, Gemini), cutting manual review effort by design.
- Applied chain-of-thought and few-shot prompting techniques to generate structured, context-aware AI feedback tailored to entry-level candidates.
- Built an NLP-powered interview simulation module with dynamic question generation and real-time AI evaluation feedback, supporting multilingual regional users.
- Architected for scalable enterprise deployment on Firebase cloud infrastructure with REST API integration across multiple LLM providers.

**Multi-PDF Semantic Search & Chatbot** | *Python · LangChain · Sentence Transformers · FAISS · ChromaDB · Streamlit · Gemini / OpenAI*

- Built a multi-document semantic search engine enabling users to upload multiple PDFs and query across all documents simultaneously using natural language questions.
- Implemented dense vector embeddings using Sentence Transformers with FAISS and ChromaDB as dual vector store options, delivering fast and accurate semantic retrieval over large document collections.
- Integrated LangChain retrieval chains with Gemini and OpenAI APIs to generate context-grounded answers, supporting queries like chapter summarization, concept explanation, and cross-document comparison.
- Deployed via Streamlit with an intuitive multi-file upload interface, making the tool accessible to non-technical users without any setup.

**PDF Q&A Chatbot (RAG)** | *Python · LangChain · ChromaDB · OpenAI API · Streamlit*

- Built an end-to-end Retrieval-Augmented Generation (RAG) chatbot enabling users to query any PDF document via natural language, reducing manual document search time significantly.
- Integrated ChromaDB vector database with LangChain for semantic chunking, embedding, and similarity retrieval, achieving accurate context-grounded answers.
- Deployed an interactive Streamlit UI allowing non-technical users to upload PDFs and receive instant AI-generated answers in a clean, responsive interface.

**Smart Traffic Signal Management System** | *Python · OpenCV · YOLOv8 · Reinforcement Learning*

- Designed a real-time Computer Vision traffic management system using YOLOv8 for multi-class vehicle detection, achieving accurate classification under varied lighting and traffic conditions.

- Applied Reinforcement Learning to adaptively optimize signal timing based on live vehicle density, demonstrating applied ML decision-making in real-world infrastructure.
- Integrated an ML-based decision engine that dynamically adjusts signal cycles, reducing average vehicle wait time through data-driven automation.

## EDUCATION

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### B.Tech — Computer Science & Engineering (AI & ML)

Expected May 2026

Hi-Tech Institute of Technology, Chh. Sambhajinagar | Maharashtra

## CERTIFICATIONS & TRAINING

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- Generative AI & Agentic AI Program — UpGrad, Hyderabad (Ongoing)
- Hands-on experience with: Claude AI, Gemini, Google AI Studio, OpenAI API, Cursor AI, Gamma AI — prompt design, workflow automation & AI content generation
- Prompt Engineering Techniques: zero-shot, few-shot, chain-of-thought, role-based prompting for optimizing LLM outputs across diverse task types

## SOFT SKILLS & LANGUAGES

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**Soft Skills:** Analytical Thinking · Rapid Learning · Cross-functional Collaboration · Effective Communication · Agile Mindset

**Languages:** English (Professional) · Hindi (Full Professional) · Marathi (Native)